Beyond Representation: Index Structure Limitations in Dense Retrieval

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Why this talk: can we drop the sparse side?

Production retrieval today is hybrid: a dense embedding index (HNSW / IVF) *plus* an inverted index (BM25). Two stores, two scoring paths, extra ops.

The tempting question: could a pure-dense stack replace hybrid? Simpler, one index, one score.

Two things have to hold:

- **Bottleneck 1 — representation.** Can the embedding *encode* what BM25 encodes? Well-studied: Tay et al. 2020, Fan et al. 2022 — capacity grows with d.
- **Bottleneck 2 — indexing.** Even if the embedding is capable, can an *ANN index* still find it? This talk. Largely unstudied.

Headline (the index alone — same embedding)

Flat $\rightarrow$ **IVF** (partition index the theory targets): forfeits $\sim 33\%$ of achievable Recall@10 on identifier queries, $< 5\%$ on standard.

Flat $\rightarrow$ **HNSW** (graph index, same population): $\sim 63\%$ on identifier queries, $< 2\%$ on standard.

The population that pays the bill: “spear-fishing” queries

Working definition

Queries whose relevance hinges on a single, low-frequency, high-IDF token: identifiers (SKUs, part numbers), rare terms, precise named entities.

- Small fraction of traffic, but operationally critical: failed spear-fishing $\Rightarrow$ empty result pages, abandoned carts, broken RAG lookups.
- BM25 handles them trivially (exact token match).
- If pure dense is going to replace hybrid, it has to hold up *here*.
- Our three query subsets: **standard** (control), **rare** (≤ 5 docs), **ID** (extreme case: doc identifier as query).

The idealized case: BM25 is embedding search

Reframe BM25 as a dense inner product in a huge, discrete embedding space $\mathbb{R}^V$ — *one dimension per token* (V = vocabulary size; for our char-trigrams, order 10^6–10^7).

Let M = number of docs in the corpus and n_t = number of docs containing token t. Each doc d becomes a sparse, IDF-weighted vector:

$$f(d)_t \;=\; \alpha_t \;=\; \log\bigl(M/n_t\bigr) \quad \text{if } t \in d, \text{ else } 0.$$

A query is a *binary indicator*: $q_t = 1$ if $t \in q$, else 0.

For a query $q = \{t^*\}$ with rare token t^*:

- Relevant doc d_{rel} (contains t^*): $\langle q, f(d_{\text{rel}})\rangle = \alpha_{t^*}$ — big, because n_{t^*} is small.
- Irrelevant doc: $\langle q, f(d)\rangle = 0$.

Perfect separation on spear-fishing. That is why BM25 works — it is nearest-neighbor search in $\mathbb{R}^V$.

Catch: V is enormous. BM25 handles it with an inverted list — ANN cannot. To use ANN we must compress $\mathbb{R}^V \to \mathbb{R}^k$ with $k \ll V$. *What compression costs is the rest of the talk.*

Compressing $\mathbb{R}^V \to \mathbb{R}^k$: Johnson–Lindenstrauss

Random linear map $A : \mathbb{R}^V \to \mathbb{R}^k$, target dimension $k \in \{256, 512, 1024\}$ (our embedding size — distinct from the document symbol d).

JL theorem (informal)

Given M vectors in $\mathbb{R}^V$, a random projection A preserves all pairwise inner products *up to a multiplicative error* ϵ with high probability, provided

$$k \gtrsim \frac{\log M}{\epsilon^2}, \qquad \text{equivalently} \qquad \epsilon \approx \sqrt{\frac{\log M}{k}}.$$

What survives, what dies:

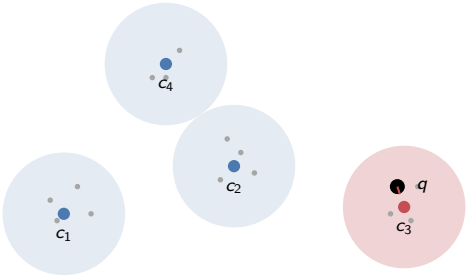
- Inner products with magnitude $\gg \epsilon$ are preserved. $\Rightarrow$ Spear-fishing signal α_{t^*} from the previous slide easily clears ϵ: **Flat search on the projected embedding is a clean oracle.**
- Magnitudes $\lesssim \epsilon$ are drowned in projection noise — indistinguishable from random.
- Bigger k shrinks ϵ; nothing else does.

So after JL alone, spear-fishing still works. The failure emerges when we add the *second* compression — clustering — on top. IVF.

Quick primer: what is an IVF index?

IVF (Inverted File) — the standard partition-based ANN index.

1. Cluster the corpus into N cells (k-means). Each cell has a *centroid* c.
2. At query time: score the query against all N centroids; keep the top n_{probe}.
3. Only search documents inside those n_{probe} cells.



Speedup: N/n_{probe} fewer distance computations. **Failure mode**: if the relevant cell's centroid is not in the top n_{probe} by query similarity, its documents are *invisible*.

The mechanism: centroid dilution (intuition)

A centroid is the *average* of the vectors in its cell.

- A *common* token appears in many docs ⇒ contributes strongly to the average ⇒ shows up in the centroid.
- A *rare* token appears in a handful of docs ⇒ its contribution is averaged *away* ⇒ nearly invisible in the centroid.

Now consider a query dominated by that rare token:

- Query vector is nonzero on the rare-token direction.
- Every centroid — *including the one containing the relevant docs* — has *almost zero* weight on that direction (averaged away).
- Result: all centroids look equally (un)related to the query. Cluster selection becomes essentially random.

The failure is not the embedding. It is the averaging step in the index.

The bound: when does the centroid signal survive?

So far: in $\mathbb{R}^V$ the relevant doc scores α_{t^*}; JL preserves it. But **IVF selects clusters by the centroid**, and a centroid is an average.

Step 1 — Centroid. Partition into N clusters, average size $|C| \approx M/N$: $c_C = \frac{1}{|C|} \sum_{d \in C} f(d)$.

Step 2 — Prevalence. Let $n_{t^*} = \#$ docs *in* C containing t^*. Only these contribute to the t^*-direction of c_C:

$$c_C\big|_{t^*} \approx \underbrace{\frac{n_{t^*}}{|C|}}_{p \;=\; \text{prevalence of } t^* \text{ in } C} \cdot \alpha_{t^*}.$$

Rare token $\Rightarrow$ p tiny (often $1/|C|$, or 0).

Step 3 — Query. $q =$ binary indicator of t^* ($q_{t^*} = 1$):

$$\boxed{\langle q,\, c_C \rangle \;\approx\; \alpha_{t^*} \cdot p}$$

Survival condition

Relevant centroid picked *only when* $\alpha_{t^*} \cdot p \gg \epsilon$ (JL floor). Rare: α big, p tiny $\Rightarrow$ product $< \epsilon \Rightarrow$ wrong cluster. Bigger k helps ϵ; nothing helps p.

HNSW isn't a way out

HNSW is not partition-based — it is a *layered proximity graph*. Greedy descent from a hub node down to the query's local neighborhood.

So why does it fail on the same queries?

- Graph edges are built from Euclidean distances during insertion.
- Dense regions of the embedding space get thick, well-connected subgraphs. Rare vectors land in sparse regions — *few incoming edges from any hub*.
- Greedy search from a hub cannot reach an orphan node without a long detour: it stops at a locally-good but globally-wrong region.
- Widening the beam (`efSearch`) helps — but with diminishing returns.

The pattern is the same: rare-token vectors are structurally hard to *reach*, whether the structure is a partition (IVF) or a graph (HNSW). Different mechanism; same victims.

Experimental setup (compact)

Datasets (1,000 queries per subset)

- Home Depot (HD): 54,667 product titles
- MS MARCO Passage v1.1: 82,360 passages

Query subsets

- standard | rare (≤ 5 docs) | ID

Representation: char-trigram + JL projection, $d \in \{256, 512, 1024\}$, unsupervised.
Indices (FAISS): Flat, IVF, HNSW (M=16, efC=200), BM25 baseline (Pyserini).

Why unsupervised trigrams?

A supervised encoder abstracts away the lexical detail that *defines* spear-fishing queries. To isolate the *index* bottleneck from the representation bottleneck we need a representation that provably preserves lexical distances (Johnson–Lindenstrauss).

Headline result: the Flat → HNSW gap

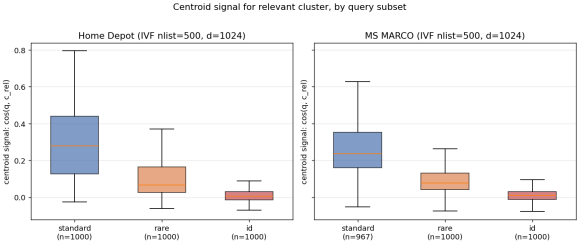
Dataset	Subset	Flat	IVF	HNSW	BM25
HD	standard	0.327	0.314	0.321	0.447
HD	rare	0.751	0.642	0.619	0.878
HD	**id**	**0.874**	0.590	**0.327**	1.000
MS MARCO	standard	0.471	0.448	0.446	0.770
MS MARCO	rare	0.205	0.187	0.166	0.688
MS MARCO	**id**	**0.174**	0.152	**0.071**	—

Recall@10 at d=1024, best ANN sweep config.

- Relative Flat→HNSW loss: HD {standard **1.8%**, rare **17.5%**, id **62.6%**}; MM analogous.
- The gap tracks the query population exactly as the theory predicts.

Direct evidence: measure the centroid signal

For each query: compute $\langle q, c_{\text{rel}} \rangle$ on the centroid of the query's *relevant* cluster (IVF, n_{list}=500, d=1024, HD).

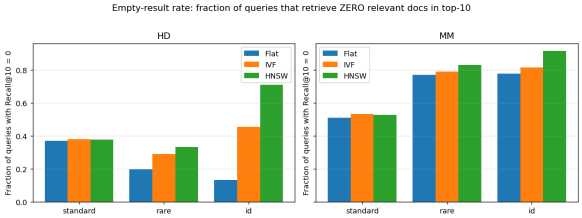


Subset	signal	rank	hit@250
standard	0.279	2	97%
rare	0.066	69	81%
id	0.007	216	59%

$\sim 40\times$ collapse of the centroid signal from standard to ID — the correct cluster is buried at rank ~ 200.

The bound $\langle c, q \rangle \approx \alpha_{t^*} \cdot p$ is not just suggestive — it is what we *measure*.

The production-visible symptom



Fraction of queries with zero relevant docs in top-10:

- HD-id: Flat **13%** → HNSW **71%**
- MM-id: Flat **78%** → HNSW **92%**
- Standard: within ±2 pp on both datasets.

Cost to match Flat recall (efSearch, HD):

- standard: **2.7× faster** than Flat
- id: **40× slower** than Flat

ANN's speedup evaporates on the population that needs it most.

“But learned encoders would fix this”

Anticipated question: *do supervised dense encoders escape the issue?*

Re-ran with `all-MiniLM-L6-v2` (Sentence-BERT, 384-dim):

Dataset	Subset	Trigram Flat R@10	**MiniLM Flat R@10**
HD	standard	0.327	0.298
HD	rare	0.751	0.428
HD	id	0.874	**0.010**
MS MARCO	standard	0.471	**0.894**
MS MARCO	id	0.174	**0.001**

- MiniLM *cannot distinguish identifiers* — “12345” $\approx$ “67890”.
- Representation collapses *before* the index has a chance to bias.
- Which is precisely why hybrid retrieval exists.

Takeaways

1. Production dense retrieval is hybrid for a reason. If you want to drop the sparse side, you have to clear *two* bottlenecks, not one.
2. Bottleneck 1 (representation) is well-studied. Bottleneck 2 (indexing) is what this paper isolates and quantifies.
3. Centroid dilution + JL noise floor give a closed-form bound; the $\sim 40\times$ centroid-signal collapse from standard to ID is exactly what the bound predicts.
4. HNSW is not a workaround — the mechanism differs, the victims don't.
5. Practical implication: query-type-aware routing, index-aware training, or — most simply — keep the sparse side.

Code & experiments: `github.com/alemagnani/sparse-dense-paper`

Thank you

Questions?

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